Time Value of Money

Chapter 4

Introduction to Financial Management (Pope)

“Time value” of Money

Suppose you must choose between the following:

1. *$10 today or $10 tomorrow*
2. *$100 today or $110 one year from now*
3. *$1 today or $2 tomorrow*
4. *$1000 today or $10000 ten years from now*

“Time value” of Money

Why…

- Inflation? Depreciating real assets? Investment opportunity?
- There is a utility cost of forgoing consumption today and you must be compensated for the disutility
- The rate of return/interest rate/discount rate captures this tradeoff

Future Value and Compounding

The value of money at the end of the stated period is called the future or compound value of that sum of money.

- Determine the attractiveness of alternative investments
- Figure out the effect of inflation on the future cost of assets, such as a car or a house.
- *"Compound interest is the eighth wonder of the world. He who understands it, earns it; he who doesn't, pays for it"*

Simple Lump-sum Deposit

Lump-sum represent a single transfer of cash over a period of time.

You make a deposit today of 100 in an account that earns 6% interest. What is the account's balance 5 years from now?

	Years					
	0	1	2	3	4	5
Interest Earned		$6.00	6.36	6.74	7.15	7.57
Value	$100.00	106.00	112.36	119.10	126.25	133.82

Future Value Formula

$$FV = PV(1 + r)^t$$

FV = Future value

PV = Present value

R = periodic interest rate

t = Time (number of periods)

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$$FV = PV(1+r)^t$$

$$FV = 100(1+0.06)^5 = \$133.82$$

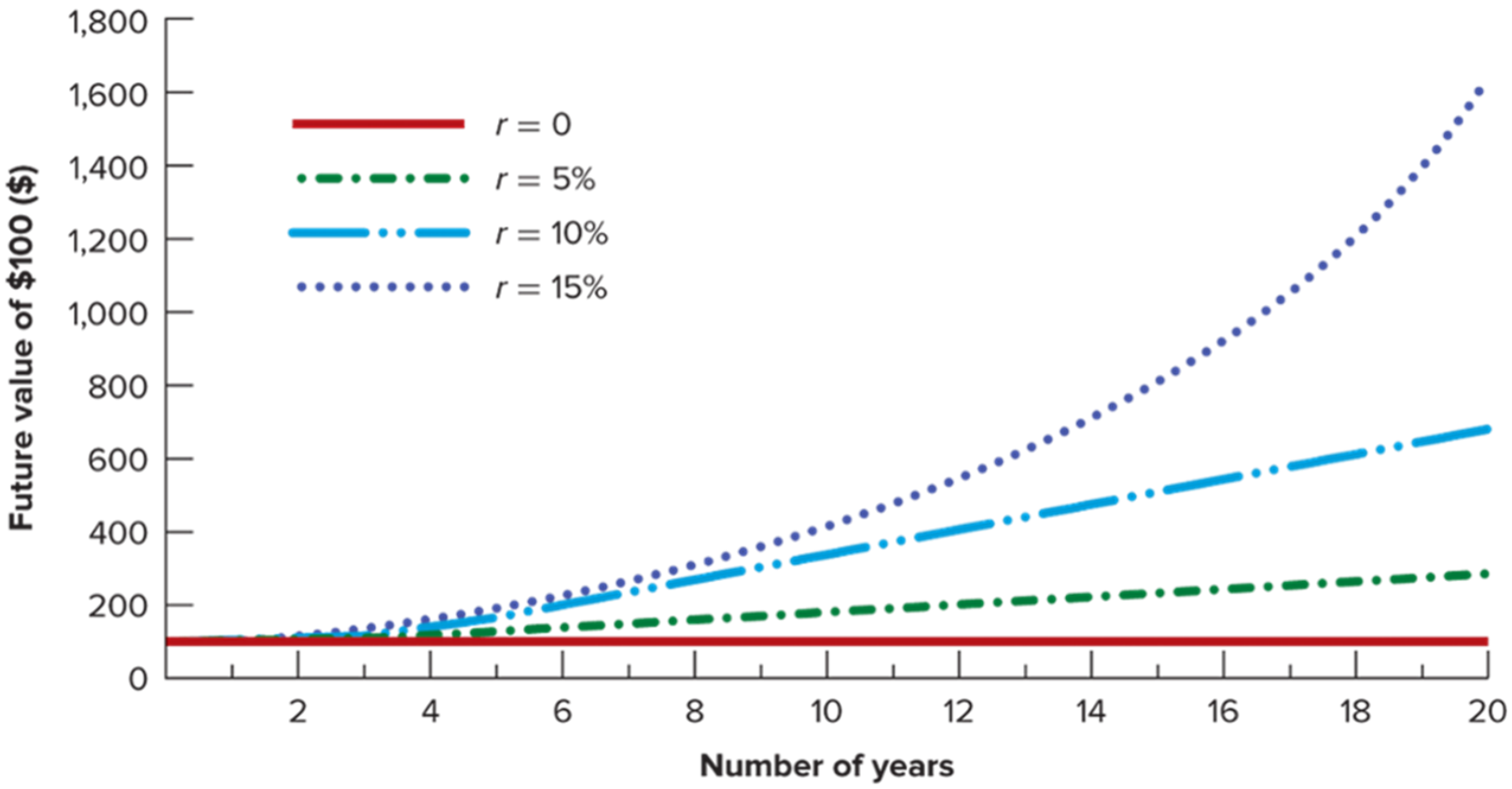
Example 1:

Let's say you want to know how much money you will have accumulated in your bank account after 4 years, if you deposit all $5,000 of your high-school graduation gifts into an account that pays a fixed interest rate of 5% per year. You leave the money untouched for all four of your college years.

Example 2:

Let's say that you have seen your dream house, which is currently listed at $300,000, but unfortunately, you are not in a position to buy it right away and will have to wait at least another 5 years before you will be able to afford it. If house values are appreciating at the average annual rate of inflation of 5%, how much will a similar house cost after 5 years?

Compound Interest Effects



Present Value and Discounting

The value of a future cash flow at the start of the stated period is called the present or discounted value of that sum

- Involves discounting the interest that would have been earned over a given period at a given interest rate
- Exact opposite of future value or compounding
- Useful in determining the value (or price) of an asset that produces cash flows in the future

Present Value Formula

$$PV = \frac{FV}{(1+r)^t} = FV \times \frac{1}{(1+r)^t}$$

FV = Future value

PV = Present value

R = periodic interest rate

t = Time (number of periods)

Example 3:

Your student loan bill of $40,000 is due in 5 years. You wish to set aside some money today in an account that earns 6% interest per year to pay this debt. How much should you set aside today to perfectly finance the student loan bill in 5 years?

One Equation and Four Variables

$$FV = PV(1+r)^t \quad \longleftrightarrow \quad PV = \frac{FV}{(1+r)^t}$$

$$\updownarrow \qquad\qquad\qquad \updownarrow$$

$$r = \left(\frac{FV}{PV}\right)^{1/t} - 1 \quad \longleftrightarrow \quad t = \left[\frac{\ln\left(\frac{FV}{PV}\right)}{\ln(1+r)}\right]$$